CYBERSECURITY HOME LAB SETUP REPORT

Submitted By

Name: Ashiga Evanjelin R

Department: B.Tech Information Technology (IT)

Date: 16 June 2026

1. Objective

The objective of this task was to build a safe and isolated cybersecurity practice environment for learning web application security testing, vulnerability assessment, and network traffic analysis.

The lab environment consists of:

- Kali Linux Virtual Machine
- OWASP Juice Shop Vulnerable Application
- Docker Container Environment
- Burp Suite
- Wireshark
- Nmap

2. Software and Tools Used

Tool	Purpose
VirtualBox	Virtualization Platform

Kali Linux	Attacker Machine
Docker	Container Management
OWASP Juice Shop	Vulnerable Web Application
Nmap	Network Scanning
Burp Suite	Web Proxy and Security Testing
Wireshark	Network Packet Analysis

3. Lab Architecture

The lab environment was configured using Kali Linux running inside VirtualBox. OWASP Juice Shop was deployed inside a Docker container hosted on Kali Linux.

Architecture:

Internet
|
VirtualBox
|
Kali Linux VM
|
Docker
|
OWASP Juice Shop (Port 3000)

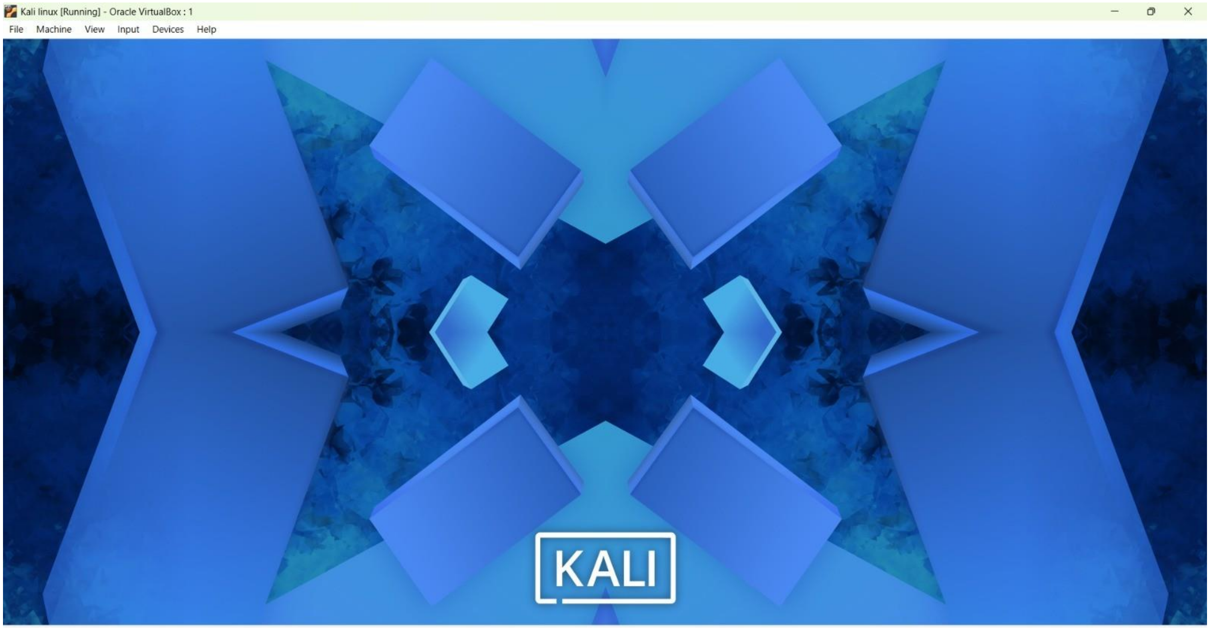
4. Kali Linux Setup

A Kali Linux virtual machine was configured in VirtualBox.

Configuration:

- Adapter 1: NAT
- Docker Service Enabled
- Security Tools Installed

Screenshot:



5. Deployment of OWASP Juice Shop

OWASP Juice Shop was deployed using Docker.

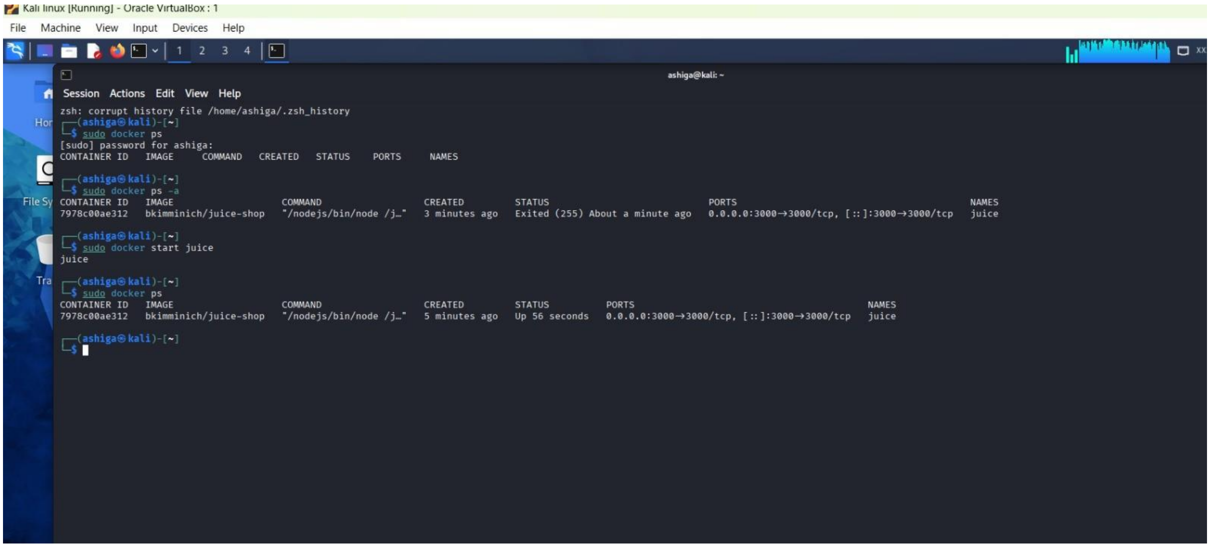
Commands Used:

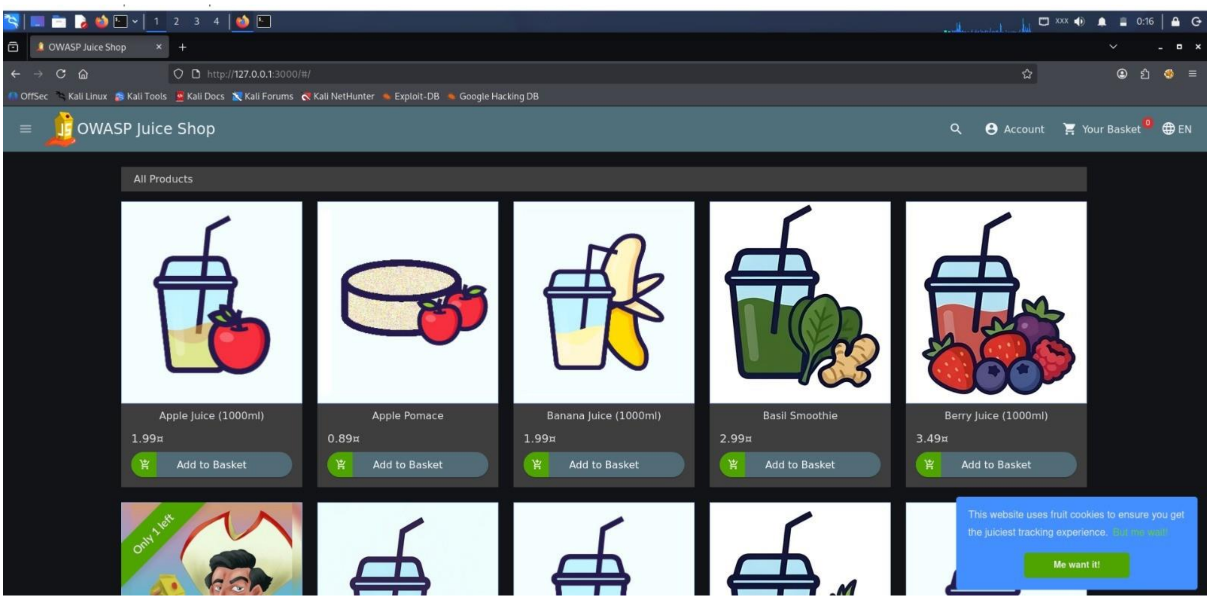
sudo docker start juice

The application was successfully started and accessed through:

http://127.0.0.1:3000

Screenshot:



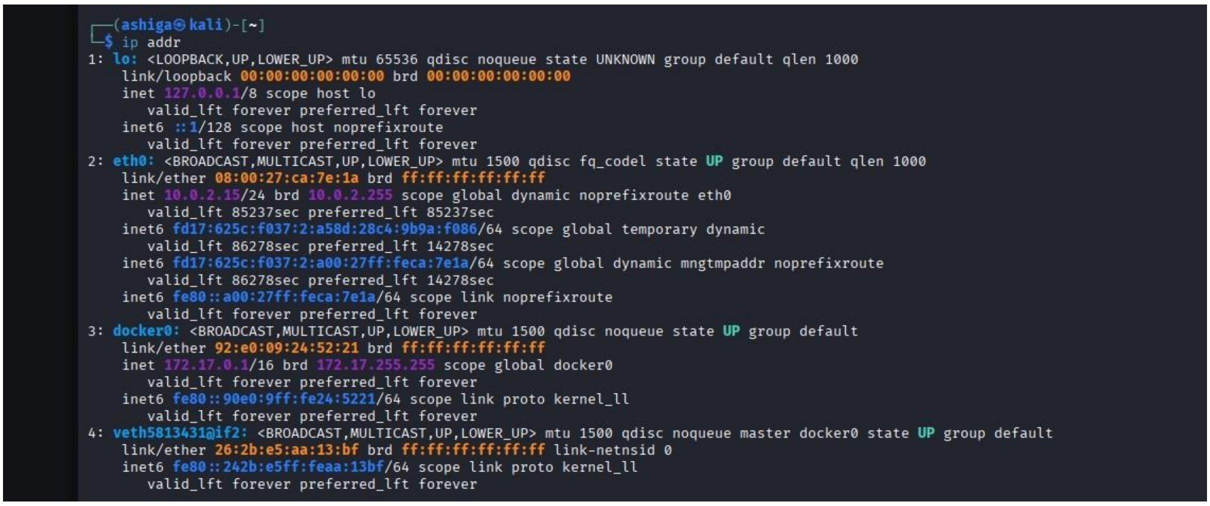


6. IP Configuration

The network configuration of the Kali Linux virtual machine was verified using the following command:

ip addr

Screenshot:



7. Network Validation Using Nmap

The Nmap tool was used to identify active services and open ports.

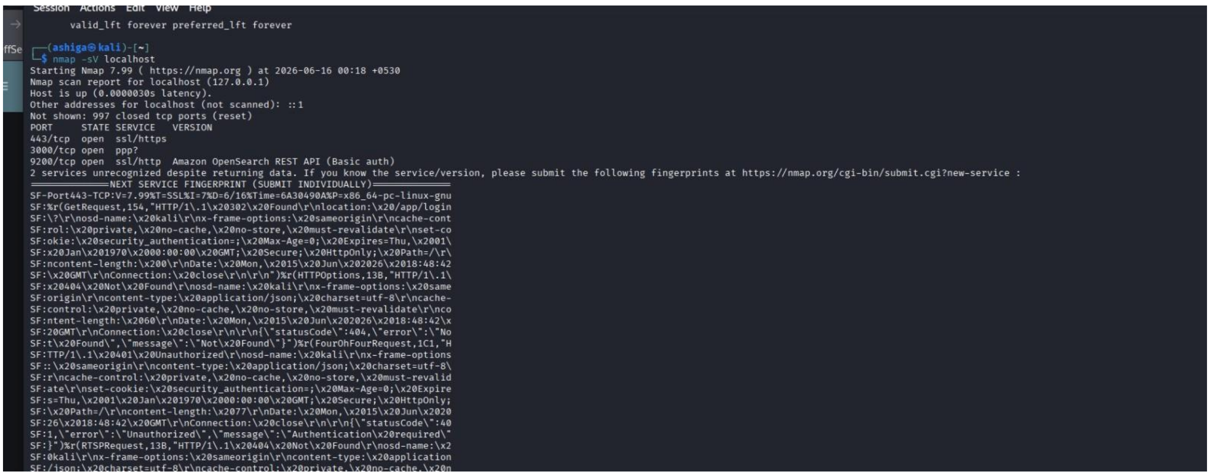
Command Used:

nmap -sV localhost

Result:

- Port 3000 detected
- OWASP Juice Shop service reachable

Screenshot:

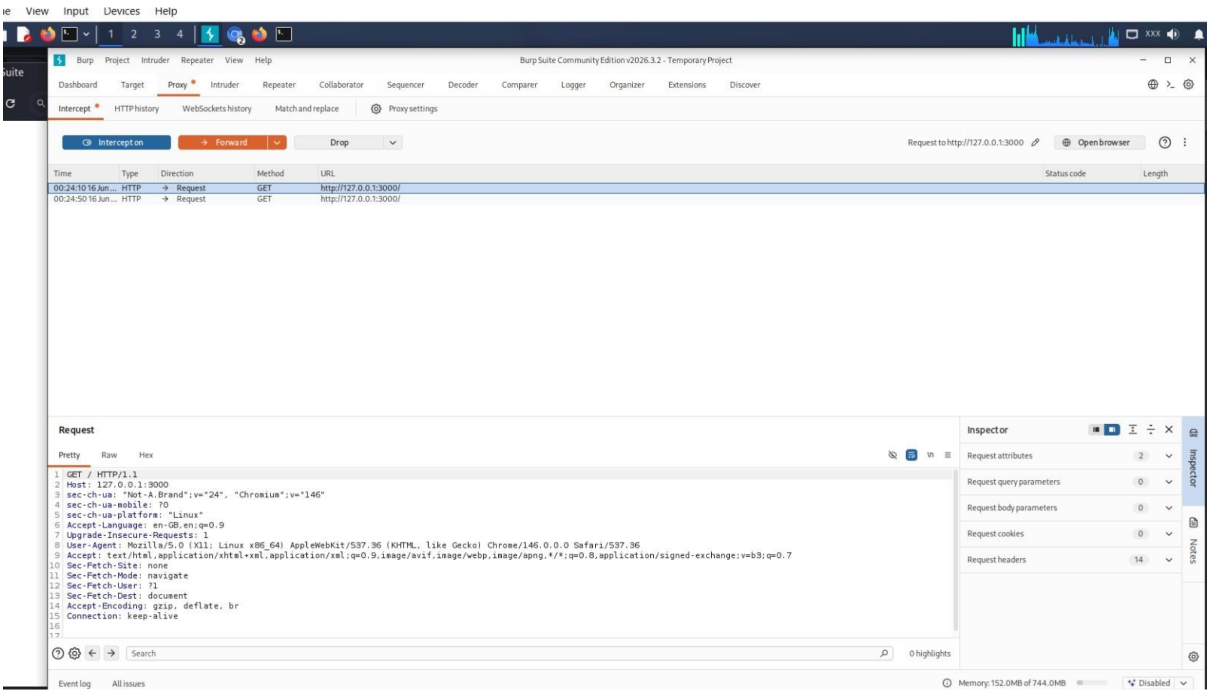


8. Burp Suite Validation

Burp Suite was launched and configured using FoxyProxy.

The Juice Shop application traffic was successfully intercepted and recorded in HTTP History.

Screenshot:



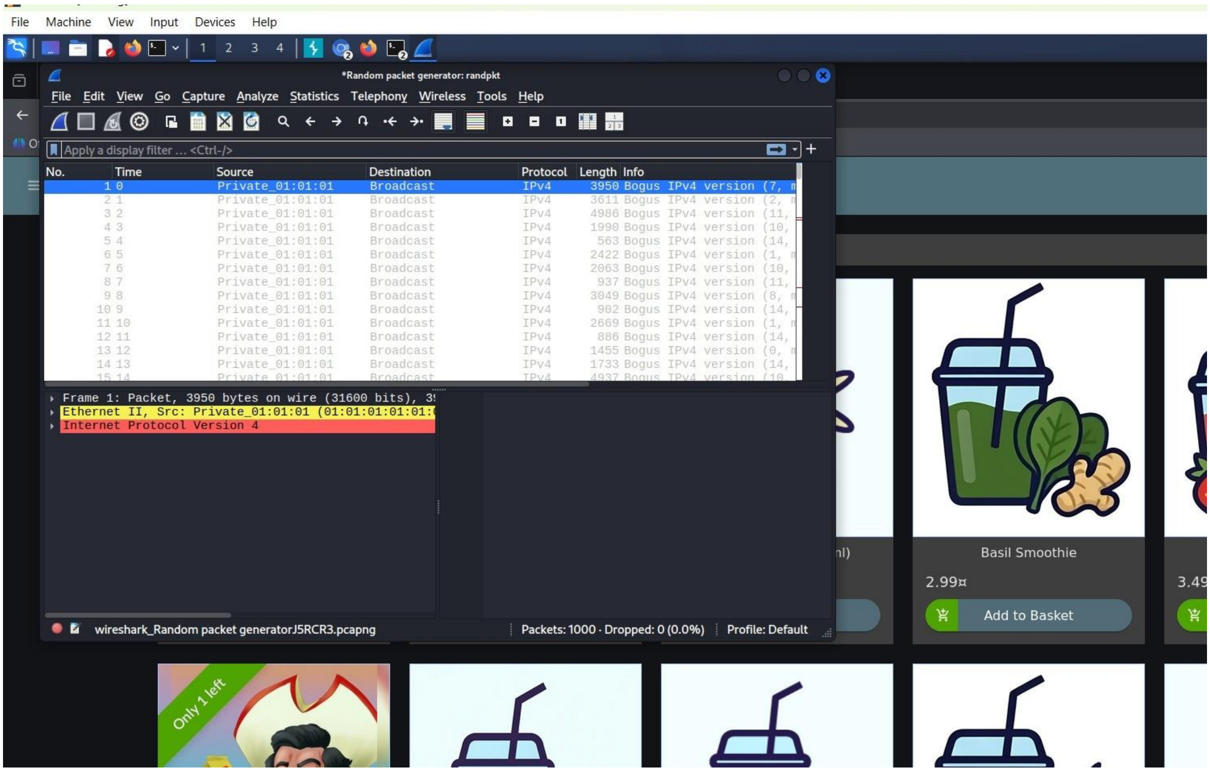
9. Wireshark Validation

Wireshark was used to capture and analyze network traffic generated during testing.

Captured Traffic:

- HTTP Requests
- Localhost Communication
- Packet Details

Screenshot:



10. Conclusion

A cybersecurity home lab environment was successfully established using Kali Linux and OWASP Juice Shop. Docker was used to deploy the vulnerable application, while Nmap, Burp Suite, and Wireshark were utilized to validate connectivity, analyze traffic, and perform security testing activities. The environment provides a safe platform for future web application security testing and cybersecurity learning exercises.